Introduction to vectors

Outcomes

A student:

- develops understanding and fluency in mathematics through exploring and connecting mathematical concepts, choosing and applying mathematical techniques to solve problems, and communicating their thinking and reasoning coherently and clearly **MAO-WM-01**
- operates with 2D and 3D vectors and uses 2D vectors to solve problems involving motion in two dimensions **ME1-12-02**

Content

Vector representation and notation

- Define a vector as a quantity having both magnitude and direction
- Associate vectors with directed line segments and recognise that a vector may have many directed line segments associated with it
- Identify and use notation for vectors in both two dimensions and three dimensions, including $\mathbf{a}$, $\underset{\sim}{a}$ and $\overrightarrow{AB}$, where $\overrightarrow{AB}$ is the vector with magnitude and direction those of the directed line segment from A to B

Example(s):

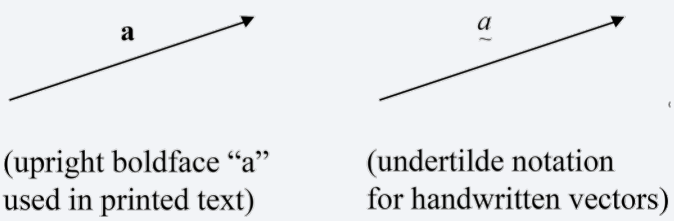

Image long description: Two identical arrows representing vectors, sloping upwards from left to right. The first is labelled with '**a**' and is captioned with 'upright boldface "a" used in printed text'. The second is labelled with '*a*' and has a tilde underneath with the caption 'undertilde notation for handwritten vectors'.

- Use notations $|\mathbf{a}|$, $\left|\underset{\sim}{a}\right|$ and $\left|\overrightarrow{AB}\right|$ to represent the magnitude of a vector
- Describe a position vector as a vector with its tail at the origin
- Represent vectors graphically with and without graphing applications
- Recognise and use the fact that two vectors are equal if they have the same magnitude and direction to solve problems

Introduction to 2D and 3D vectors

- Use Cartesian coordinates to represent points in 2-dimensional (2D) and 3-dimensional (3D) space with and without graphing applications
- Use the midpoint and distance formulas in two dimensions and three dimensions

- Identify that, in three dimensions, all points on the xy-plane have a z-coordinate of 0, and deduce similar properties for points on the xz and yz-planes and thus the equations of the three coordinate planes
- Define the zero vector $\mathbf{0}$, written as $\underset{\sim}{0}$, as the vector with zero magnitude and no direction
- Define unit vectors as vectors of magnitude 1
- Recognise and use $\hat{\mathbf{a}}$ and $\underset{\sim}{\hat{a}}$ as the notation for the unit vector in the direction of $\mathbf{a}$
- Define the standard perpendicular unit vectors $\mathbf{i}$ and $\mathbf{j}$ in two dimensions and $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ in three dimensions
- Express 2D vectors in component form, $x\mathbf{i} + y\mathbf{j}$; as an ordered pair, (x, y); and in column vector notation $\begin{pmatrix} x \\ y \end{pmatrix}$
- Recognise $\overrightarrow{AB} = \begin{pmatrix} u - x \\ v - y \end{pmatrix}$ as the vector associated with the directed line segment from the point $A(x, y)$ to $B(u, v)$ in two dimensions
- Express 3D vectors in component form, $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$; as an ordered triple, (x, y, z); and in column vector notation $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$
- Recognise $\overrightarrow{AB} = \begin{pmatrix} u - x \\ v - y \\ w - z \end{pmatrix}$ as the vector associated with the directed line segment from $A(x, y, z)$ to $B(u, v, w)$ in three dimensions

Operating with vectors

- Define a scalar as a real number that is used to multiply a vector
- Represent geometrically a scalar multiple of a vector in two dimensions and three dimensions with and without graphing applications
- Perform multiplication of a vector by a scalar algebraically in component form
- Establish and identify $\mathbf{a} = k\mathbf{b}$, for a non-zero scalar k, as a condition for two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ to be parallel to each other and determine with justification if two vectors are parallel to one another
- Identify $\begin{pmatrix} b \\ -a \end{pmatrix}$ and $\begin{pmatrix} -b \\ a \end{pmatrix}$ as vectors perpendicular to $\begin{pmatrix} a \\ b \end{pmatrix}$ and with equal magnitude
- Perform addition and subtraction of vectors algebraically in component form, and verify, with and without graphing applications, that geometrically these are obtained using the triangle law or the parallelogram law
- Establish and calculate the magnitude of a vector using $|x\mathbf{i} + y\mathbf{j}| = \sqrt{x^2 + y^2}$ for 2D vectors and $|x\mathbf{i} + y\mathbf{j} + z\mathbf{k}| = \sqrt{x^2 + y^2 + z^2}$ for 3D vectors
- Use the magnitude of a vector to find the unit vector $\hat{\mathbf{a}} = \frac{1}{|\mathbf{a}|}\mathbf{a}$ in two dimensions and three dimensions

Further operations with vectors

- Define $\mathbf{a} \cdot \mathbf{b} = x_1x_2 + y_1y_2$ as the scalar (dot) product of vectors $\mathbf{a} = x_1\mathbf{i} + y_1\mathbf{j}$ and $\mathbf{b} = x_2\mathbf{i} + y_2\mathbf{j}$ and use the scalar product to solve problems
- Define $\mathbf{a} \cdot \mathbf{b} = x_1x_2 + y_1y_2 + z_1z_2$ as the scalar product of vectors $\mathbf{a} = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$ and $\mathbf{b} = x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}$ and use the scalar product to solve problems
- Use $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$ as a geometric expression of the scalar product of non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ in two dimensions and three dimensions, where θ is the angle between the vectors and $0° \leq \theta \leq 180°$.

- Verify the equivalence of $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$ with the algebraic definition of the scalar product, $\mathbf{a} \cdot \mathbf{b} = x_1x_2 + y_1y_2$ for two dimensions and $\mathbf{a} \cdot \mathbf{b} = x_1x_2 + y_1y_2 + z_1z_2$ for three dimensions

Example(s):

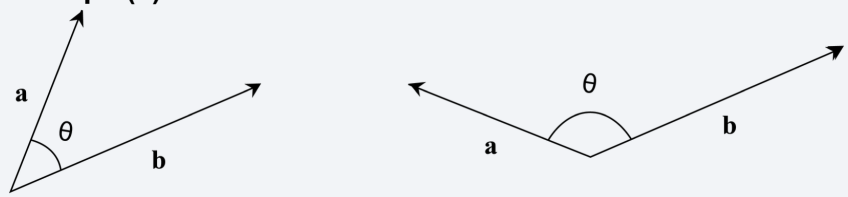

Image long description: Two vector diagrams, each represented by two arrows connected at a point. The first shows the arrows labelled a and b with an acute angle labelled theta between them. The second shows the arrows labelled a and b with an obtuse angle labelled theta between them.

- Derive and use the property $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$ to establish the scalar product definition of the magnitude of a vector ($|\mathbf{a}| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$) in two dimensions and three dimensions
- Calculate the angle between two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$, in both two dimensions and three dimensions, using the scalar product by deriving and applying the relationship $\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = (\hat{\mathbf{a}} \cdot \hat{\mathbf{b}})$
- Establish $\mathbf{a} \cdot \mathbf{b} = 0$ as a condition for two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ to be perpendicular to each other and use it to determine if two vectors are perpendicular
- Establish $\mathbf{a} \cdot \mathbf{b} = \pm|\mathbf{a}||\mathbf{b}|$ as another way to determine if two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel
- Define the projection of a vector $\mathbf{a}$ onto a vector $\mathbf{b}$, denoted by $\text{proj}_{\mathbf{b}}\mathbf{a}$, to be the vector component of $\mathbf{a}$ in the direction of vector $\mathbf{b}$

Example(s):

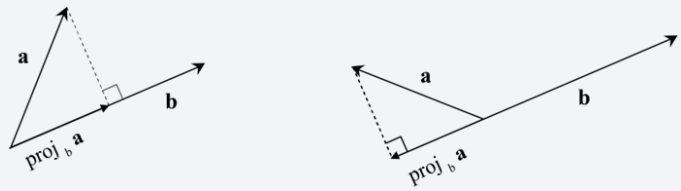

Image long description: Two vector diagrams both representing vectors, each showing two arrows connected at a point. In the first diagram, the angle between the arrows is acute. One arrow is labelled with boldface a, the other with boldface b, with the b arrow as the longer vector, pointing up and to the right. A dotted line perpendicular to arrow b joins the tip of arrow a. There is an arrow drawn on top of arrow b from the vertex, which is labelled proj b a, with a subscripted b and boldface a. In the second diagram the angle between the arrows is obtuse. One arrow is labelled with boldface a and the other with boldface b, with b as the longer vector, pointing up and to the right. A dotted line perpendicular to arrow b joins the tip of arrow a, pointing up and to the left, to an extension of arrow b, pointing down and to the left. The arrow drawn on top of arrow b from the vertex is labelled proj b a with a subscripted b and boldface a.

- Examine the proof of the formula $\text{proj}_{\mathbf{b}}\mathbf{a} = \left(\frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{b}|^2}\right)\mathbf{b} = (\mathbf{a} \cdot \hat{\mathbf{b}})\hat{\mathbf{b}}$ and use the formula to solve problems

- Determine that the component of a vector $\mathbf{a}$ perpendicular to another vector $\mathbf{b}$ is $\mathbf{a} - \text{proj}_{\mathbf{b}}\mathbf{a}$

Example(s):

Define and find the vector component of **a** perpendicular to **b**.

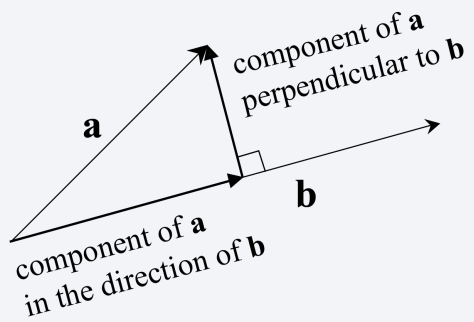

Image long description: Two arrows representing vectors, connected at a point. The angle between the arrows is acute. The arrows are labelled with boldface a and boldface b with b as the longer vector. An arrow, perpendicular to arrow b and starting from arrow b, joins the tip of arrow a and is labelled 'component of (boldface) a perpendicular to (boldface) b'. Another arrow is drawn on top of arrow b from the vertex to the point where the perpendicular arrow starts and is labelled 'component of (boldface) a in the direction of (boldface) b'.

Motion in vector form in two dimensions

- Describe the position of an object at a point in 2D space using a vector
- Describe the changing positions of an object by expressing its vector as a function of time using $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, or $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ where x and y are functions of time t
- Recognise that $x(t)$ and $y(t)$ form a pair of parametric equations for the path of the object
- Find the Cartesian equation of the path of an object, where the path is a straight line, parabola or circle
- Express the change in an object's position between two points as a displacement vector and recognise the magnitude of the displacement vector as the distance between the two points
- Solve motion problems involving constant velocity using vectors
- Solve relative velocity problems involving constant crosswind/cross-current using vector diagrams, and describe the direction of a vector where required
- Find the velocity vector $\mathbf{v} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j}$ and acceleration vector $\mathbf{a} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j}$ of an object using differential calculus
- Find the position vector and the velocity vector of an object using integral calculus given its acceleration vector
- Solve motion problems involving non-constant velocity using vectors

Projectile motion

- Recognise that the gravitational force on a mass may be regarded as a constant acting in a downwards direction when the motion of the object is restricted to a small region near the Earth's surface
- Model and analyse a projectile's path where the projectile is a point and air resistance is negligible, subject to only acceleration due to gravity, assuming that the projectile is moving close to the Earth's surface
- Represent the motion of a projectile using vectors

- Recognise that the horizontal and vertical components of the motion of a projectile can be represented by horizontal and vertical vectors
- Derive and use the equations of motion of a projectile in vector form by splitting 2D motion into horizontal and vertical components to solve problems on projectiles
- Find the Cartesian equation of the path of a projectile using parametric equations for the horizontal and vertical components of the displacement vector
- Determine features of the flight of a projectile, including time of flight, maximum height, range, instantaneous velocity and impact velocity
- Solve problems relating to the path of a projectile in which the initial velocity and/or angle of projection may be unknown, in a variety of contexts